

02 Secret Messages

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Write a program that can be used to exchange confidential messages. Your program should be able to do the following:

1. Given a key, encrypt plaintext.
2. Given a key, decrypt ciphertext.

You can use any programming language you wish, but you must provide instructions so that I can run your program.

Please submit the following:

1. Your source code
2. Instructions on how to run your program
3. An encrypted message that I can decrypt, and the key and any other information that I need
4. Answers to the following questions
 - a. What cipher algorithm did you use? What mode? Why?
 - b. What key size are you using? Why?
 - c. How will users exchange keys? Explain your reasoning.
 - d. How did you generate the key you gave me?
 - e. What could go wrong? How could your program be compromised?

Hint

Most (all) of you are comfortable with Python, in which case you will probably use the `pyca/cryptography` package. Documentation is here: <https://cryptography.io/en/latest/>